



CS 206: Computer Networks

Course Code: CS 206

Course Name: Computer Networks

L-T-P-C: 3-0-0-3

Prerequisites: None

Approval: 5th Senate

Course Contents

- **Part 1:** Building a network; Goals; Requirements; layered network architecture; Physical media - electrical, optical, wireless; Encoding; Framing; Network adapter; Error Detection; Reliable transmission
- **Part 2:** Topology: Ethernet, Rings; Packet Switching; Forwarding; Bridges and LAN switches; Wifi
- **Part 3:** Routing; Algorithms (Distance Vector; Link State); Addressing; IPv4 and IPv6; Internet; Multi-cast; [8 Lectures]
- **Part 4:** Transport protocols UDP, TCP; Network resource allocation Flow and Congestion Control; Quality of Service [8 Lectures]
- **Part 5:** Application layer - Web, Email, DNS; Network security; Network management; Advanced topics: Wireless networks; Optical communications [8 Lectures]

Note: Architectures, protocols, implementation and performance to be covered in all parts.

Textbooks:

1. Peterson & Davie, **Computer Networks: A Systems Approach**
2. A.S. Tanenbaum, **Computer Networks**.